



StockSense

Every move, accounted for

Connected inventory management



Ayush Pandey / Team X

The warehouse handoff

Planned quantities and physical work need a shared record.

10

planned units

8

reported units

**Who
approves?**

The manager needs the quantity, the reason, and the person responsible.

One workspace for every role

Administrator

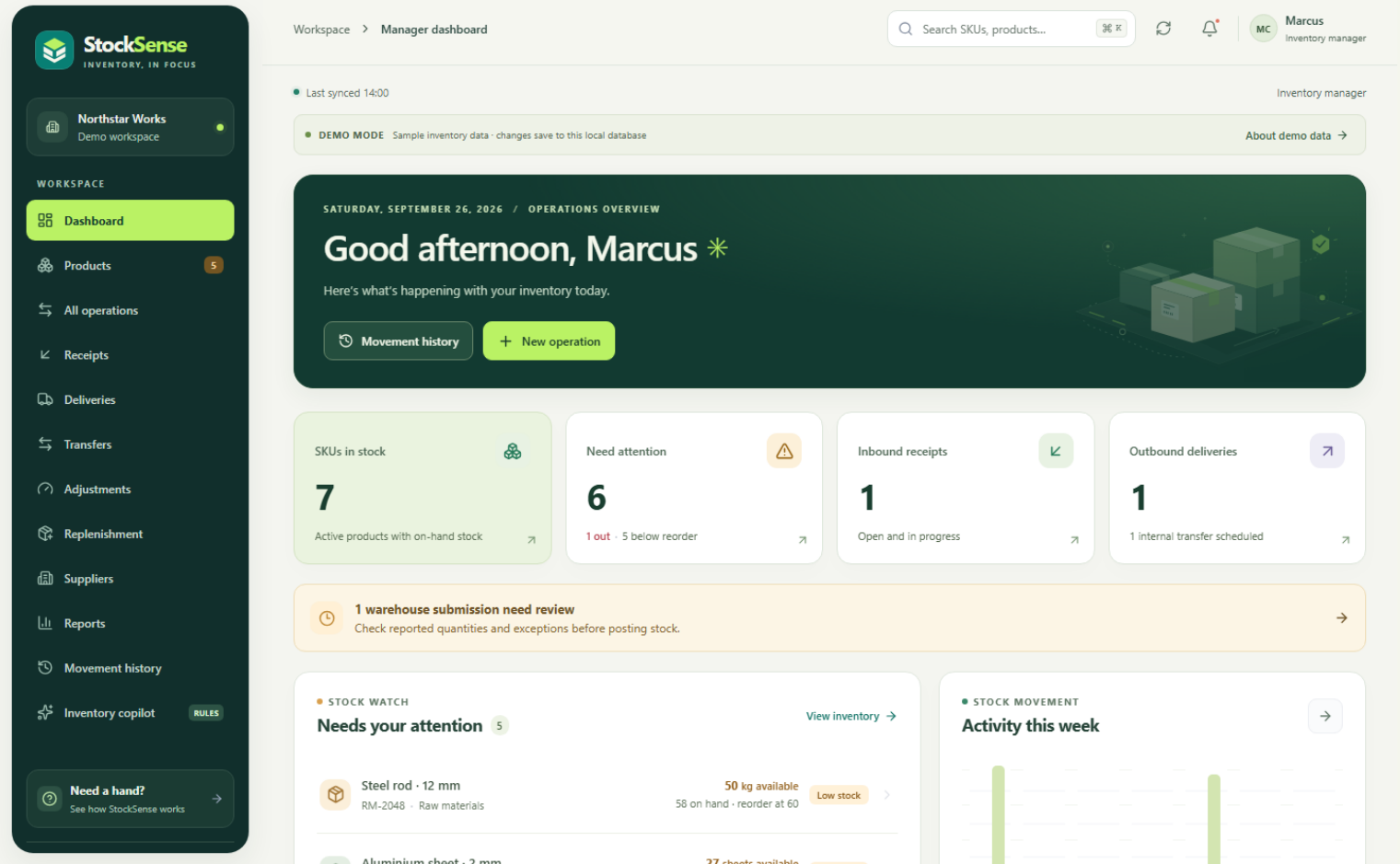
Access and warehouse assignments

Inventory manager

Planning, exceptions and posting

Warehouse staff

Assigned work and actual quantities



A controlled posting sequence

01 Plan the movement

Drafts record intent without changing stock.

02 Stage the work

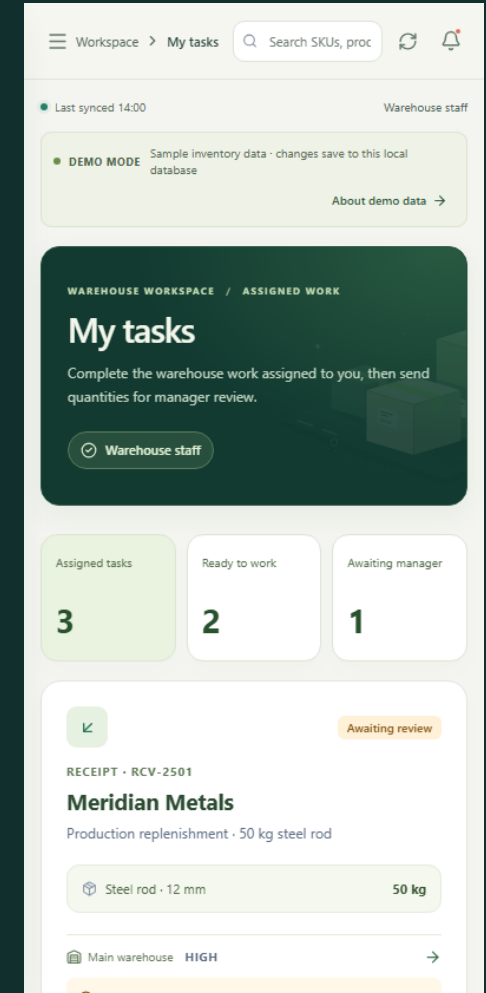
Ready deliveries and transfers reserve stock.

03 Report what happened

Assigned staff submit quantities and exceptions.

04 Review and post

Manager validation updates balances and ledger.



One handoff, end to end

A receipt moves through staff review and manager approval.

Operation details

RCV-2513

RECEIPT / RCV-2513

Meridian Metals

External supplier → Main warehouse

Draft

Ready

Posted

The draft is staged and can be validated by a manager.

Products in this operation

QUANTITY / PROGRESS

Steel rod - 12 mm

RM-2048 - kg

0 / 24

planned

Created by

Marcus Chen

Assigned to

Sam Rivera

Priority

HIGH

Created

Sep 26

Expected

Sep 26

Last updated

Just now

Receiving check: confirm all 24 kg are in good condition and submit actual quantity for manager review.

WAREHOUSE SUBMISSION

Awaiting manager review

Sam Rivera - Just now

Steel rod - 12 mm

24 kg

24 kg received, counted and checked. No damage found.

Return with instructions

Explain what the staff member should check or correct

Request changes

Review quantities to post

This action updates live stock.

Steel rod - 12 mm

24 kg remaining

24

kg

This posts a permanent, traceable ledger entry.

Confirm and post

46 kg

before the receipt

46 kg

awaiting review

70 kg

after manager approval

Captured demo: RCV-2513, 24 kg received into Main warehouse.

The stock record has safeguards

Safe retries

The same request key reuses its result.
Conflicting quantities are rejected.

Balanced transfers

Source and destination entries
commit in the same transaction.

Fresh counts

Intervening movements invalidate
a stale physical count.

Protected reservations

A count cannot reduce on-hand stock
below active commitments.

Answers grounded in inventory

Stock facts first

Availability, reorder points and recent movements inform the answer.

Rules work locally

Optional AI explanations.
Human approval for posting.

The image displays the StockSense Inventory Copilot interface, which is designed to provide AI-powered inventory management insights. It consists of a mobile application (left) and a web dashboard (right).

Mobile App Interface:

- Header:** StockSense INVENTORY, IN FOCUS
- Workspace:** Northstar Works (Demo workspace)
- Workspace Menu:** Dashboard, Products, All operations, Receipts, Deliveries, Transfers, Adjustments, Replenishment, Suppliers, Reports, Movement history.
- Inventory copilot:** A button labeled "RULES" is visible.
- User Profile:** Marcus Chen, Inventory manager.

Web Dashboard Interface:

- Header:** Workspace > Inventory copilot
- Search:** Search SKUs, products...
- Status:** Last synced 14:00
- Mode:** DEMO MODE (Sample inventory data - changes save to this local database)
- Section:** STOCKSENSE INTELLIGENCE / GROUNDED IN YOUR DATA
- Inventory copilot:** A button labeled "Rule-based mode" is visible.
- Ask your inventory:** A section for asking questions about inventory, with a note: "Answers use current company stock and posted delivery history. No stock changes happen here."
- TRY ASKING:** A list of suggested questions: "What may run out this month?", "Why is steel rod below its reorder point?", "What should I reorder next?", "Show me items with too little history to forecast".
- StockSense Copilot:** A section for asking questions about inventory, with a note: "Current inventory - deterministic rules".
- Question:** "Which products are low on stock?"
- Answer:** "Aluminium sheet - 2 mm: 27 sheets on hand, 0 sheets reserved, and 27 sheets available against a 36 sheets reorder point. There is not enough recent delivery history for a reliable usage forecast; the rule-based top-up to the threshold is 9 sheets. Nitrile gloves - M: 138 boxes on hand, 0 boxes reserved, and 138 boxes available against a 140 boxes reorder point. There is not enough recent delivery history for a reliable usage forecast; the rule-based top-up to the threshold is 2 boxes. Safety glasses - clear: 0 pcs on hand, 0 pcs reserved, and 0 pcs available against a 18 pcs reorder point. There is not enough recent delivery history for a reliable usage forecast; the rule-based top-up to the threshold is 18 pcs."
- CURRENT EVIDENCE:** A table showing inventory data for Aluminium sheet - 2 mm and Nitrile gloves - M.
- Aluminium sheet - 2 mm:** On hand 27 sheets, Reserved 0 sheets, Reorder point 36 sheets, Lead time 14 days, Insufficient usage history. Main warehouse 23 on hand - 23 available, Production floor 4 on hand - 4 available. Rule-based top-up: 9 sheets.
- Nitrile gloves - M:** On hand 138 boxes, Reserved 0 boxes, Reorder point 140 boxes, Lead time 8 days, Insufficient usage history. Main warehouse 126 on hand - 126 available, Production floor 12 on hand - 12 available. Rule-based top-up: 2 boxes.
- Footer:** "Responses are suggestions. Verify stock and review every operation before posting." and "Answers are grounded in your current workspace data".

Built and checked

27

security regression groups

14

production scenarios

13

browser scenarios

React + TypeScript / Express / SQLite

Workflow smoke tests, build and dependency audit also checked.

Local automated checks. Scope and remaining limits appear in the verification report.

A pilot with measurable outcomes

Proposed first users

Small warehouse teams with frequent stock handoffs

Measure	Pilot question
Reconciliation time	Can the team close a count faster?
Exception turnaround	How quickly can a manager resolve a mismatch?
Movement traceability	Can each posted quantity link to its operation?

Pilot proposal. No adoption, savings or customer results are claimed.

The next milestones

01 **Validate the pilot**

Measure value and willingness to pay with a warehouse team.

02 **Improve floor workflows**

Barcode capture, richer reporting and larger catalogue support.

03 **Prepare the deployment**

TLS, restore drills, monitoring and a plan beyond one database node.



Every posted movement is traceable

StockSense

Ayush Pandey / Team X / Contact X



Architecture

Layer	Technology	Responsibility
Client	React + TypeScript	Role screens, forms and accessible interactions
API	Express + Zod	Authorization, validation and stock transactions
Storage	SQLite	Balances, operations, ledger, audit and sessions
Optional providers	OpenAI / Resend	Explanations and reset-email delivery

Current deployment model: one application node and one SQLite database.

Questions worth asking

How is this different?

The staff-to-manager handoff connects physical work to approved stock changes.

What if AI is unavailable?

Labeled inventory rules remain available. Posting still requires a person.

How would it earn revenue?

Proposed workspace subscriptions by locations and active users.

What needs work before rollout?

TLS, restore drills and monitoring.
Scale work beyond SQLite on one node.